PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [redacted]
Tracking No. m8b-0kc7-tqh0
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-6964
Comment on FR Doc # 2025-02305

Submitter Information

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Organization: Arizona State University

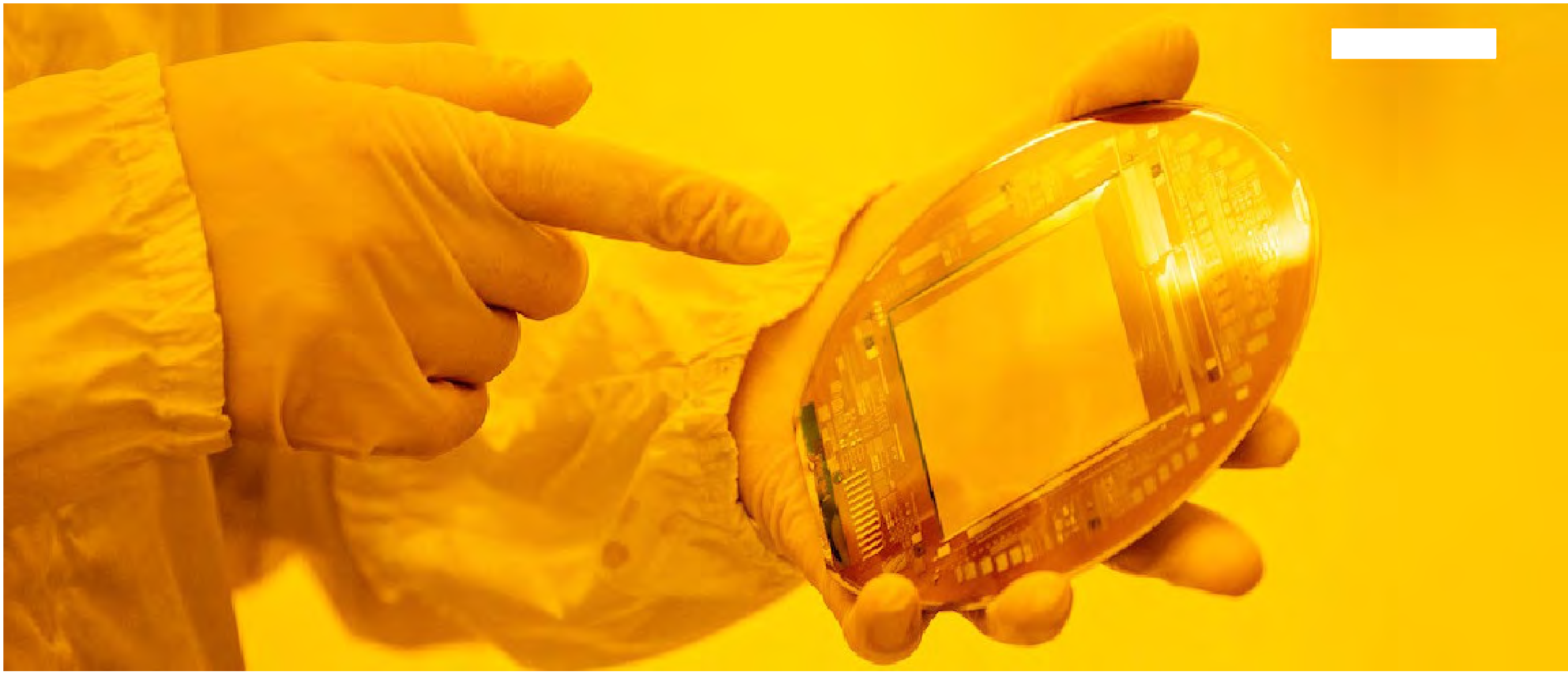
General Comment

See attachment.

Attachments

ASU Response to OSTP AI Action Plan (3-15-25)

ASU Response to AI Action Plan RFI (3-15-25)



Arizona State University

Request for Information on the Development of an Artificial Intelligence Action Plan

March 15, 2024

Arizona State University

Executive Summary

Arizona State University (ASU) is pleased to submit the following materials in response to the request for information from the Networking Information and Technology Research and Development (NITRD) Program's National Coordination Office (NCO) on behalf of the Office of Science and Technology Policy (OSTP).

ASU's comments to inform the development of the Artificial Intelligence (AI) Action Plan ("Plan") draw on our expertise as a leading public research university committed to U.S. dominance in AI and related technologies. ASU is a comprehensive public research institution prioritizing national service at scale, with total enrollment reaching 181,000 students in Fall 2024 across physical campuses and extensive online programs.

Complementing ASU's significant educational reach is a robust research enterprise, with research expenditures totaling nearly $1 billion. ASU has also consistently been recognized as a leader in innovation and ranked the most innovative school in the United States by U.S. News & World Report for ten consecutive years.

Below are answers to the questions posed in the request for public comment and additional background on ASU's programs and capabilities aligned with the Plan.

Responses

Education and Workforce

Developing engineering and AI-ready talent at scale is critical to continued U.S. leadership. ASU is educating the AI workforce of the future with the necessary skills to lead in both engineering for AI and AI for engineering. Scaling requires a multi-pronged approach to workforce development, including degree programs, non-degree programs, K-12 engagement, and faculty development. ASU also offers upskilling and reskilling programs to help fill national workforce gaps in AI-related skills among current employees.

For degree-seeking students, ASU's Fulton Schools of Engineering represent the largest and one of the most comprehensive engineering programs in the nation, with more than 33,000 students enrolled in Fall 2024. To achieve this scale, ASU developed and deployed new education technologies that have enabled the University to quadruple the number of STEM graduates over the last two decades and triple the number of engineering students over the last decade — from 9,300 to more than 30,000 students. And ASU will continue scaling its program to support U.S. AI leadership.

For example, ASU established a new School of Manufacturing Systems and Networks in 2021, building upon the University's existing manufacturing and systems engineering programs while strategically centering its curriculum, research portfolio, and industry engagements to meet the growing demands of an evolving manufacturing landscape in semiconductors and related technologies. A primary purpose of this new school is to help fill the more than 2 million manufacturing jobs projected to be created by 2030, all of which AI will enable.

Additionally, public research universities and community colleges must work together to expand the AI-ready workforce, including everything from hardware manufacturing to infrastructure construction and maintenance to integrating AI into every sector of the economy.

Policy Recommendations: We recommend the Plan prioritize helping every American become AI-ready through access to educational and workforce development programs, including non-degree upskilling and reskilling programs. A national initiative of this scale requires government universities, community colleges, and industry all working together. To that end, the Plan should:

- Support the creation, expansion, and standardization of AI degree programs, certifications, and technical training.
- Provide incentives for companies to create internship and apprenticeship programs that give students and workers hands-on experience.
- Incentivize companies to invest in AI workforce training through university and community college partnerships.
- Establish regional AI training centers where higher education institutions, industry, and federal, state, and local governments collaborate to develop the local workforce.
- Establish an AI Workforce Task Force to convene industry, higher education institutions, and federal agencies to align workforce development efforts with economic and national security priorities.

Application and Use

ASU is applying AI to advance research, improve learning, and prepare students to be lifelong learners. In January 2024, ASU became the first university to partner with OpenAI to leverage and scale its AI capabilities for education. ASU's partnership with OpenAI has focused on enhancing student success, forging new avenues for innovative research, and expanding access to cutting-edge AI tools for students and researchers. This partnership helps faculty generate innovative research proposals on AI applications across various fields, including education, energy, supply chains, and healthcare. The Plan should continue encouraging academic-industry partnerships to use AI to solve the most pressing societal challenges across education, national security, energy, and health.

Policy Recommendation: We recommend the Plan prioritize the application of AI to solve the nation's most pressing challenges. Such solutions could include developing advanced AI-enabled education technologies.

Explainability and Assurance of AI Model Outputs

ASU researchers are advancing verification technology to detect AI-generated content, including identifying voice cloning used in widespread scams. For example, ASU's OriginStory project uses new microphone technology to verify that a human speaker produced the recorded speech, then watermarks the speech as authentically human. The watermark can be shown to listeners, establishing trust from recording to retrieval. This verification work will protect Americans from scams and could be used to verify the authenticity of evidence for criminal prosecutions.

Policy Recommendation: We recommend the Plan prioritize programs for AI verification technology generated by industry and academia to protect Americans from scams and other criminal activity.

National Security, Defense, and Cybersecurity

AI-driven cyberattacks pose a serious challenge to America's cybersecurity. As AI technologies become more accessible, even hackers with minimal technical skills will be more likely to execute widespread, sophisticated attacks on critical infrastructure successfully. At the same time, defenders can use AI to improve cybersecurity. AI-powered cybersecurity tools leveraging enhanced pattern-detection capabilities are especially promising for identifying cybersecurity threats, as are emerging approaches to applying formal methods to cybersecurity.

However, cybersecurity is not the only national security domain in which AI will have a significant impact. New approaches to human, AI, and robot teaming offer capabilities that neither humans nor robots can achieve independently. For example, ASU's Center for Human, Artificial Intelligence, and Robot Teaming (CHART) combines research across computer science, robotics, law, art, and social science to create novel human-machine systems through teaming. CHART's interdisciplinary offers a model for further enhancing the capabilities of humans and machines by working together.

Policy Recommendation: We recommend the Plan prioritize R&D for cybersecurity use-cases. We also recommend emphasizing human-machine teaming, especially applying human-machine teaming capabilities to national security.

AI Hardware

Universities play a central role in developing next-generation semiconductors essential to fielding increasingly advanced AI systems. In partnership with industry, universities can accelerate the transition from research to prototyping to mass production of AI hardware.

Policy Recommendation: We recommend the Plan continue to support domestic AI hardware R&D, prototyping, and scaling by academia and industry.

Data Centers

The rapid adoption of AI tools across sectors has vastly increased demand for data centers. To meet this need, ASU works with utilities and industry partners to design next-generation data centers with enhanced computing power while improving operational efficiency by reducing energy and water usage.

Energy consumption and energy costs are especially significant issues for AI data centers. University research can help address specific challenges in future data center design, development, and operation, including energy resilience, AI-driven optimization, and advanced materials and construction techniques.

AI tools can also help to manage shifting data center power demand by dynamically balancing and redistributing computational workloads across regions, increasing resiliency, and mitigating risks associated with natural disasters and national security-related disruptions.

Finally, atmospheric water harvesting is a method of water collection that draws water from humidity in the air. Such techniques can reduce data centers' demand for water, aiming to make next-generation data centers net-zero water consumers.

<u>Developing Non-Water-Cooling Techniques</u>. R&D into novel coolants other than water and new approaches to the built environment, like integrated air-cooling infrastructure, also present pathways for reducing data center water demand.

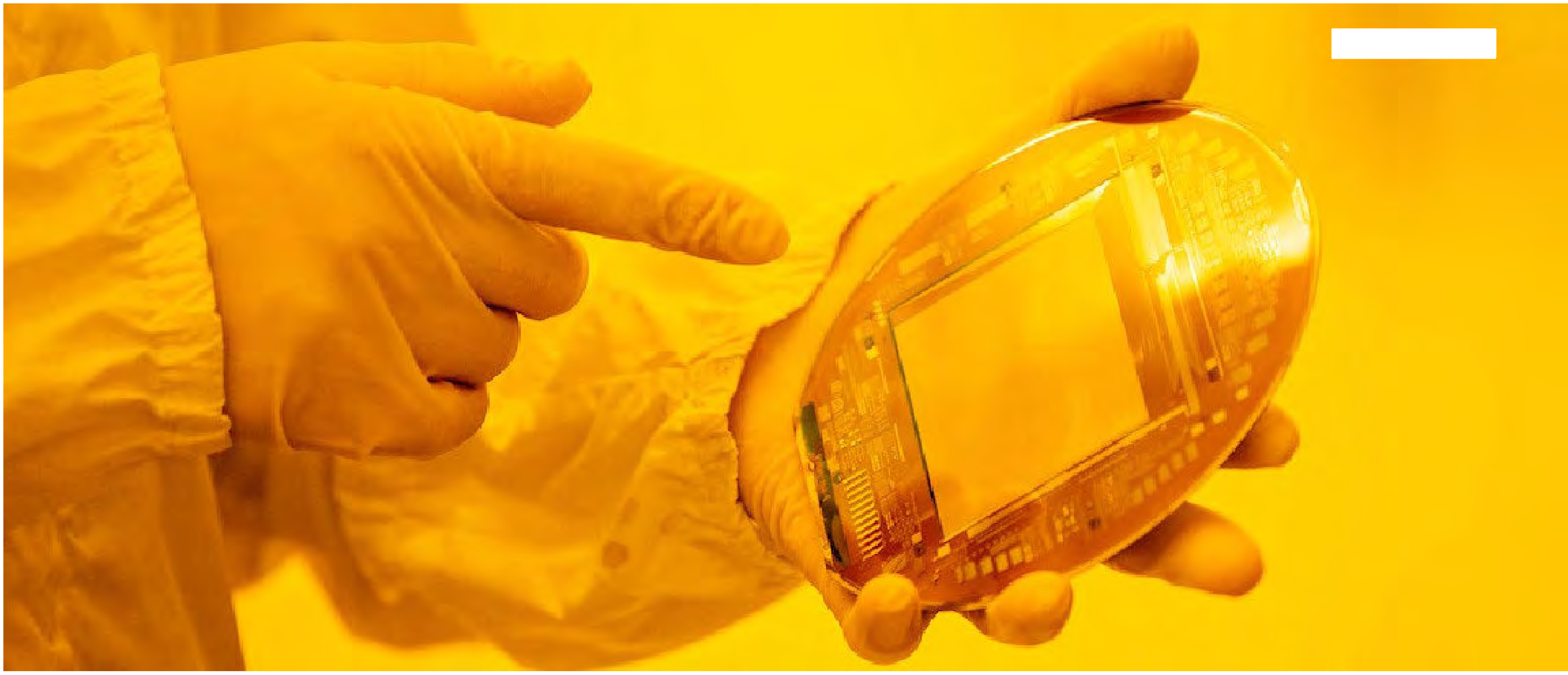
<u>Policy Recommendation</u>: We recommend that the AI Action Plan support university and industry R&D on AI-powered workload distribution, predictive maintenance, and automated resource management to enhance data centers' capabilities and operational efficiency. We recommend the Plan prioritize funding for universities to research and deploy new energy and water efficiency systems for AI data centers.

Open-Source Development and the National AI Research Resource

Open-source AI tools advance research, foster innovation, and ensure equitable access to cutting-edge technology within universities and educational settings. Open-source AI is essential for university research and training the next generation of AI practitioners.

Similarly, ASU supports the objectives of the National AI Research Resource (NAIRR), which include making more advanced computing clusters available to academic researchers.

<u>Policy Recommendation</u>: We recommend that the Plan encourage the development of open-source models – especially for academia -- while ensuring safeguards to protect intellectual property and national security interests. We also recommend continuing to scale the NAIRR.



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